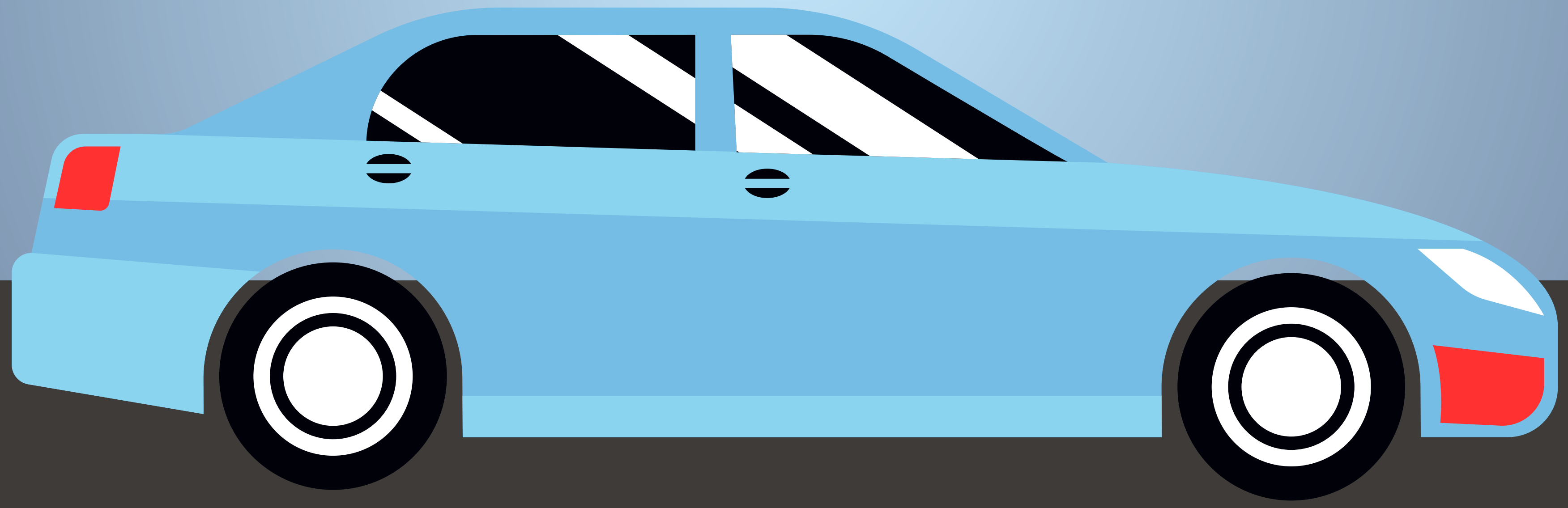
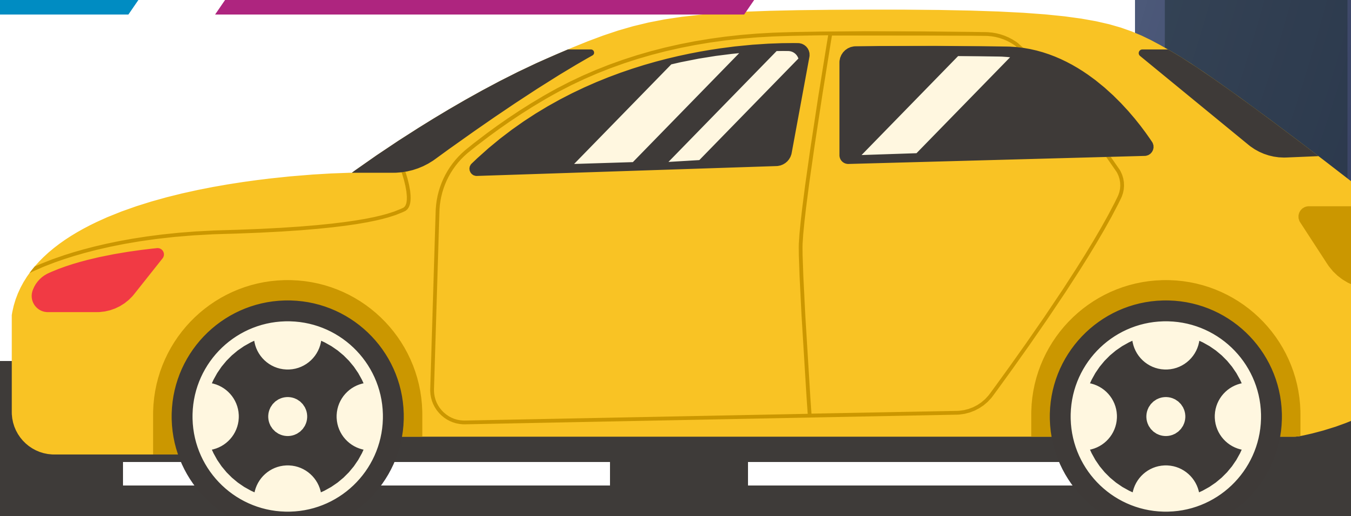
DATA-DRIVEN INSIGHTS ON PREDICTING CAUSE OF CRASH OUTCOMES IN CHICAGO



Project by Cynthia Jemutai

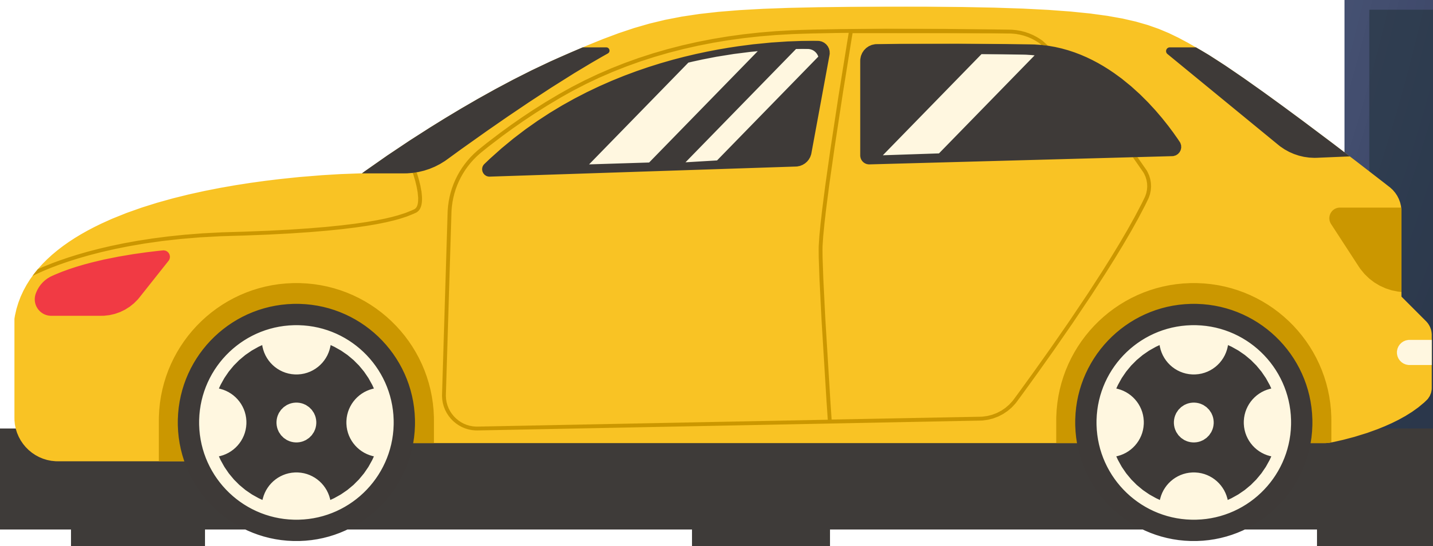
OVERVIEW

- BUSINESS AND DATA UNDERSTANDING
- DATA ANALYSIS
- MODELLING
- EVALUATION
- RECOMMENDATION



BUSINESS PROBLEM

- Road accidents are a major safety and economic concern in Chicago
- Many crashes are influenced by:
  - Driver behavior(maneuver)
  - Vehicle type
  - Number of passengers
- Our goal: use data to understand how crashes happen and how to prevent them



DATA UNDERSTANDING

Dataset: Traffic Crashes Vehicles

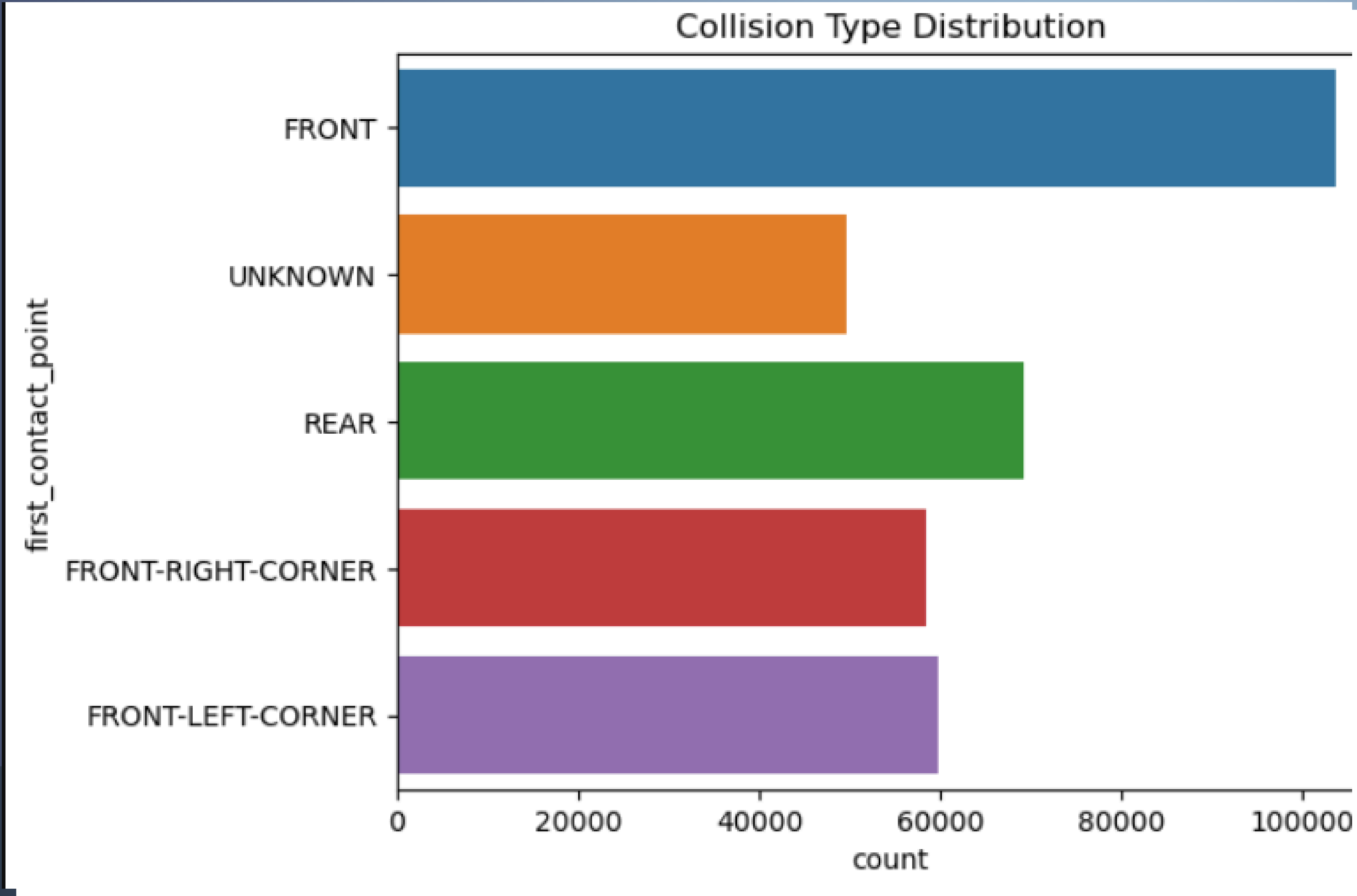
Key variables:

- Vehicle type
- Number of passengers
- Occupant count
- Driving maneuver (what the driver was doing)
- First point of contact (how the crash happened)

Key Insight:

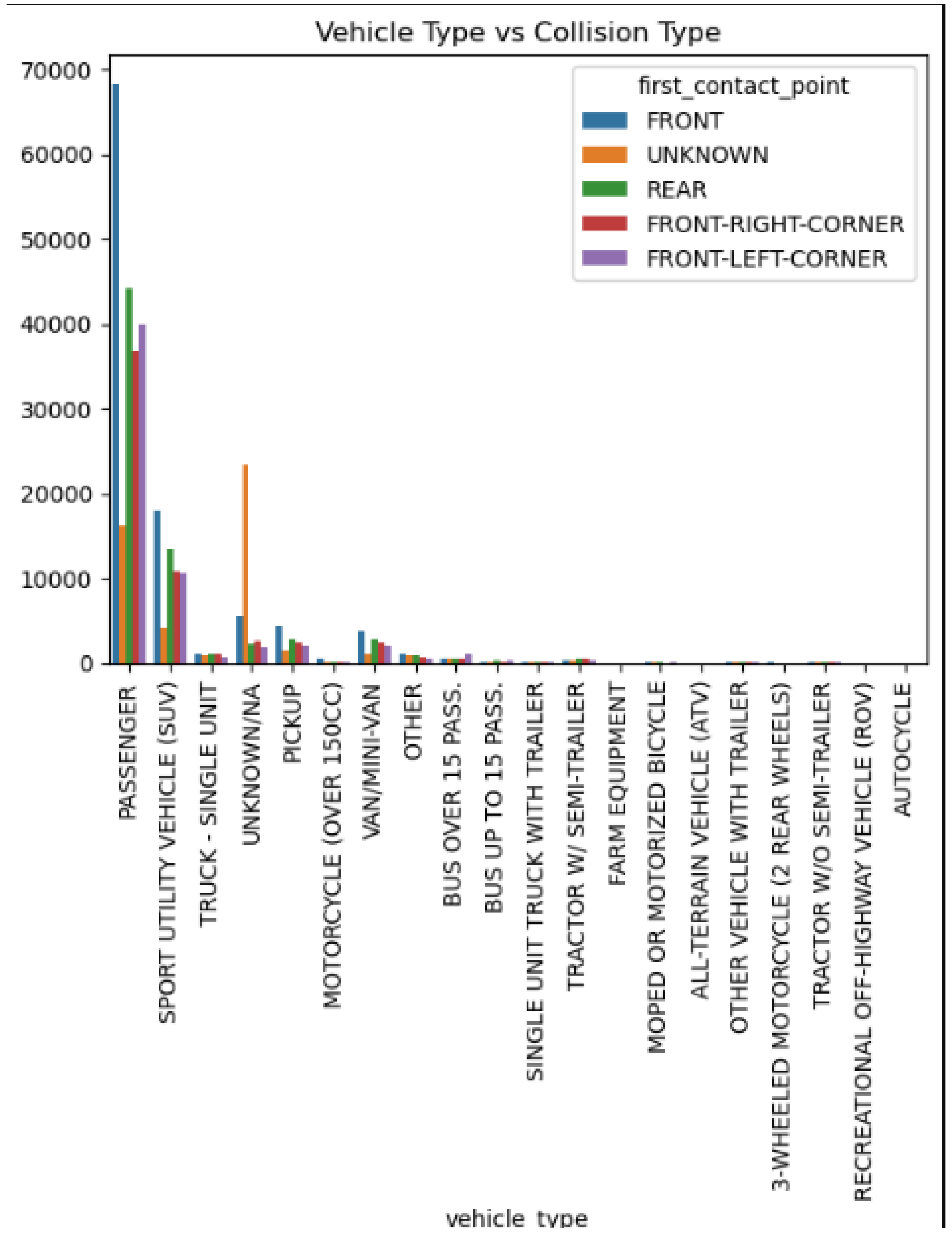
- We used collision type as a proxy for crash cause

DATA ANALYSIS & VISUALIZATION

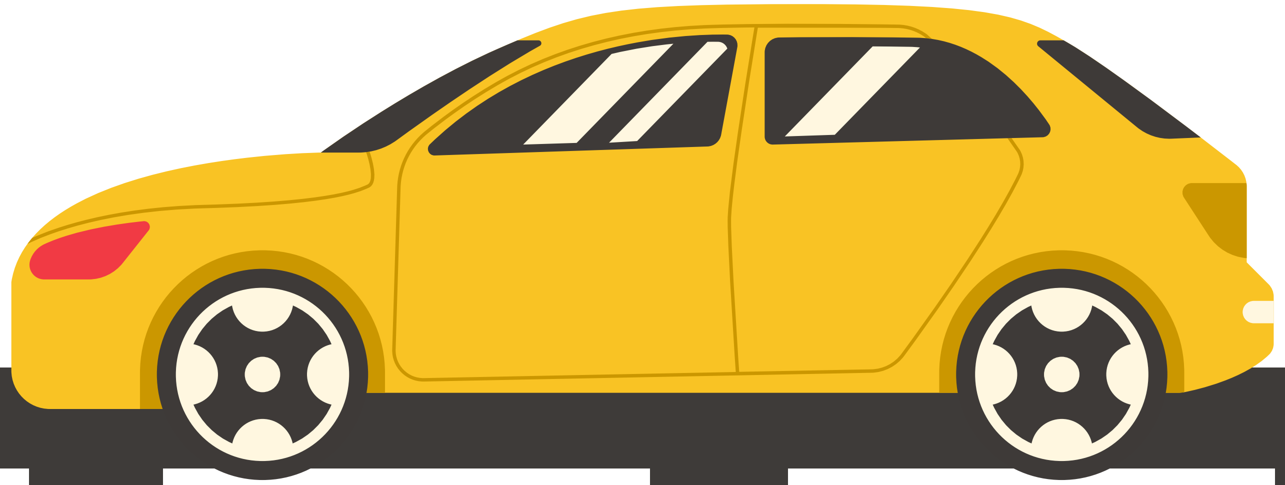


The graph shows that collusions from the front are more common

DATA VISUALISATION



Passanger's vehicle types are involved in front end contact crash patterns



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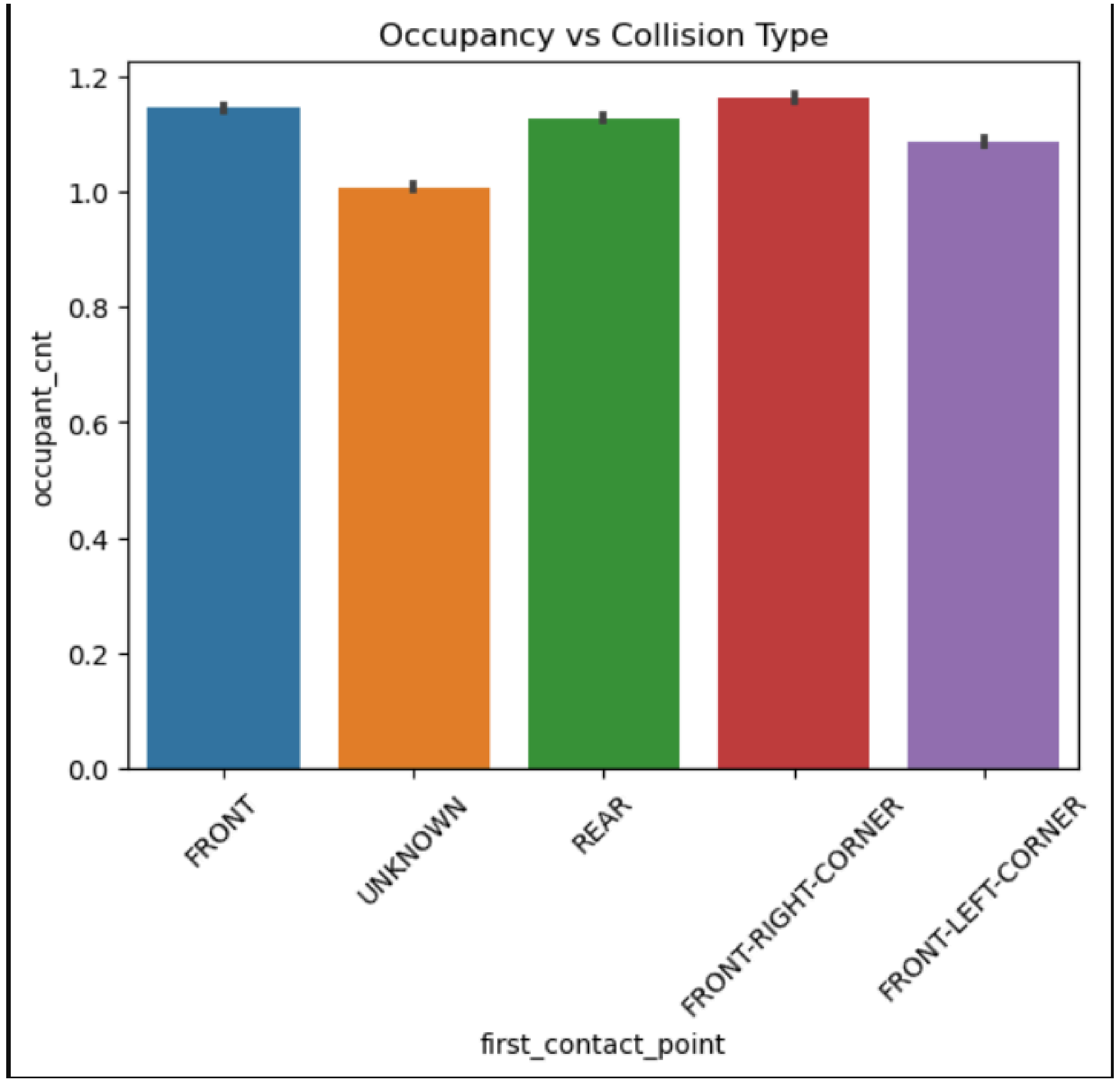
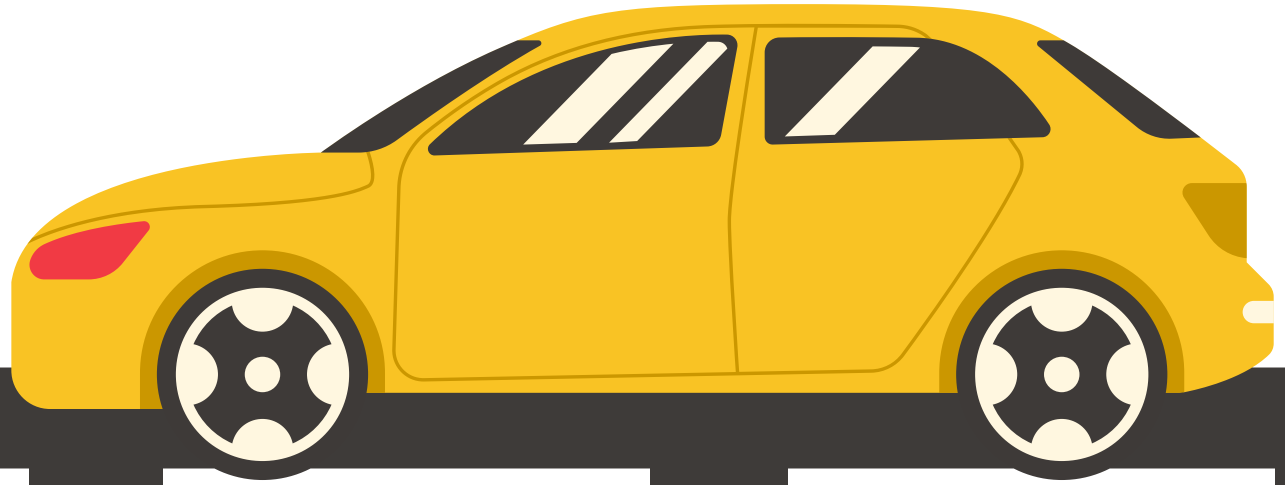


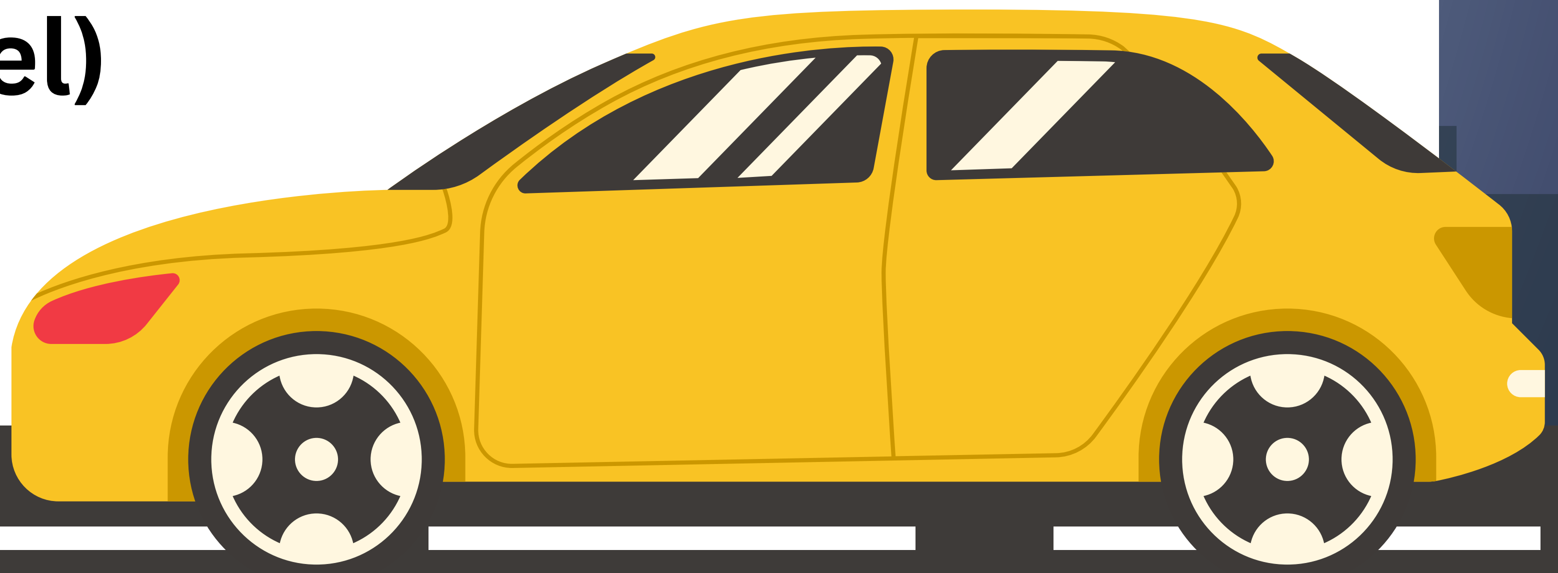
Passenger occupancy affects crash outcomes
we see that Higher occupancy slightly linked to the front



MODELLING

We trained a machine learning model to:
Predict how a crash happens
With the following Steps:

1. **Cleaned and prepared the data**
2. **Converted categories into numbers**
3. **Trained two models:**
    - **Logistic Regression (simple baseline)**
    - **Random Forest (advanced model)**



RECOMMENDATION

1. **Enhance Driver Behavior Interventions**
    - **Focus on high-risk maneuvers such as turning and sudden stopping**
2. **Vehicle-Specific Safety Policies**
    - **Encourage adoption of advanced safety features such as collision avoidance systems**
3. **Occupancy-Based Safety Measures**
    - **Strengthen enforcement of occupancy-related regulations**


THANK
YOU
For Your Attention !